

E-Commerce Customer Churn & Customer Lifetime Value (LTV) Optimization

Project Report

Prepared By

Name: Aravindh S

Role: Data Analyst Portfolio Project

Tools Used:

- Microsoft Excel
- MySQL
- Python
- Jupyter Notebook
- Power BI

1. Project Overview

Customer churn is one of the major challenges faced by e-commerce businesses. Losing existing customers directly impacts revenue and long-term business growth. This project aims to analyze customer behavior, identify high-risk customers, calculate Customer Lifetime Value (LTV), and provide business insights that help improve customer retention.

The project was developed using Excel, SQL, Python, and Power BI to demonstrate a complete end-to-end Data Analytics workflow.

2. Problem Statement

The company wants to understand:

- Which customers are likely to churn.
- Which customers generate the highest revenue.
- How customer behavior affects churn.
- Which membership groups are most loyal.
- How to improve customer retention.
- How Customer Lifetime Value can be increased.

3. Objectives

The primary objectives of this project are:

- Analyze customer purchasing behavior.

- Identify high-risk customers.
- Measure Customer Lifetime Value.
- Analyze customer engagement.
- Build interactive dashboards.
- Generate business recommendations.

4. Dataset Information

A realistic e-commerce customer dataset was generated using Python and the Faker library.

Dataset Size

- Total Records: 50,000
- Total Columns: 25

Important Columns

- Customer_ID
- Gender
- Full_Name
- Age
- Email
- Phone_Number
- City
- State
- Join_Date
- Membership_Type
- Total_Orders
- Average_Order_Value
- Total_Spend
- Days_Since_Last_Purchase
- Login_Count
- Coupon_Usage
- Support_Calls
- Return_Count
- Refund_Amount

- Preferred_Category
- Loyalty_Score
- Customer_Lifetime_Value
- Customer_Satisfaction
- Risk_Level
- Churn

5. Data Cleaning

The dataset was cleaned before analysis.

The following tasks were performed:

- Removed duplicate records.
- Checked missing values.
- Verified unique Customer IDs.
- Standardized data types.
- Corrected phone number format.
- Ensured realistic customer information.
- Verified Customer Lifetime Value.
- Ensured Refund Amount never exceeded Total Spend.

6. Excel Analysis

Excel was used for:

- Data validation
- Duplicate checking
- Missing value analysis
- Pivot Tables
- Pivot Charts
- Basic business analysis

Pivot Tables created:

- Customers by Membership Type
- Churn by Membership
- Revenue by Category

- Customer Distribution by State

7. SQL Analysis

SQL was used for querying customer information and business insights.

Operations performed include:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- Aggregate Functions
- CASE Statements
- Subqueries
- Common Table Expressions (CTE)
- Window Functions
- Ranking Functions
- LEAD()
- LAG()

Example business questions answered:

- Customers spending above average.
- Top loyal customers.
- High-risk customers.
- Average spend by membership.
- Customer ranking.
- Churn analysis.

8. Python Analysis

Python was used for:

- Dataset generation.
- Data preprocessing.
- Exploratory Data Analysis (EDA).

- Statistical analysis.
- Feature engineering.
- Customer Lifetime Value calculation.
- Churn business logic generation.

Libraries Used

- Pandas
- NumPy
- Faker
- Matplotlib

Python operations:

- Data Cleaning
- Missing Value Check
- Duplicate Removal
- Histograms
- Correlation Analysis
- Summary Statistics

9. Power BI Dashboard

An interactive dashboard was developed in Power BI.

Dashboard Pages

Dashboard 1

Revenue Analysis

KPIs

- Total Revenue
- Average Order Value
- Total Customers
- Total Orders

Charts

- Revenue by Category
- Revenue by Membership
- Revenue Trend

- Top Categories

Dashboard 2

Customer Churn Analysis

KPIs

- Churn Customers
- Retained Customers
- Churn Rate
- Average Customer Lifetime Value

Charts

- Churn by Membership
- Churn by State
- Churn by Category
- Risk Level Distribution

Slicers

- State
- Membership
- Category
- Risk Level

10. Key Business Insights

The analysis revealed several important business insights.

- Customers with higher login activity showed lower churn probability.
- Higher loyalty scores were associated with increased Customer Lifetime Value.
- Customers with frequent support calls were more likely to churn.
- Bronze membership customers showed higher churn rates compared to Gold and Platinum members.
- Customers with higher spending generated significantly greater lifetime value.
- High refund amounts negatively impacted customer profitability.

11. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Provide loyalty rewards to high-value customers.
- Improve customer support response time.
- Offer personalized promotions to high-risk customers.
- Encourage frequent customer engagement.
- Improve membership upgrade programs.
- Reduce product return rates.
- Improve customer satisfaction through better service.

12. Challenges Faced

During the project, several challenges were encountered:

- Creating realistic customer data.
- Maintaining business logic.
- Handling duplicate records.
- Data cleaning.
- Designing an effective dashboard.
- Creating SQL business queries.
- Generating meaningful customer insights.

13. Skills Demonstrated

Technical Skills

- Excel
- MySQL
- Python
- Power BI
- Data Cleaning
- Data Visualization
- Exploratory Data Analysis
- Business Intelligence

Soft Skills

- Problem Solving

- Analytical Thinking
- Data Interpretation
- Business Understanding
- Dashboard Design

14. Conclusion

This project successfully demonstrates an end-to-end Data Analytics workflow. Starting from data generation and cleaning, the project progressed through SQL analysis, Python-based exploratory data analysis, and interactive Power BI dashboard development. The analysis identified customer behavior patterns, churn risk, and customer lifetime value, enabling meaningful business recommendations.

The project showcases practical skills in Excel, SQL, Python, and Power BI while emphasizing real-world business problem solving. It can serve as a portfolio project demonstrating the ability to perform data preparation, analysis, visualization, and business insight generation.

15. Future Enhancements

The project can be further enhanced by:

- Building a Machine Learning model for customer churn prediction.
- Developing a real-time Power BI dashboard.
- Connecting the dashboard to a live SQL database.
- Integrating cloud storage and automated data refresh.
- Deploying the dashboard on the Power BI Service.
- Adding predictive analytics and customer segmentation models.

16. References

- Microsoft Excel Documentation
- MySQL Documentation
- Python Documentation
- Pandas Documentation
- NumPy Documentation
- Faker Documentation
- Power BI Documentation